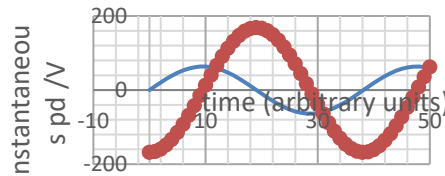


Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)	Pd of L and C are opposite signs (antiphase) (or in phasor diag) (1) Pd of L 90° before $\frac{R}{I}$ and C 90° after $\frac{R}{I}$ (or in phasor diag) (1) Pd of supply is the resultant of all 3 (or in phasor diag) (1)	3			3		
		(ii)	Resonance: R pd is power supply pd (1) Pds of L and C are equal and opposite (1)	2			2		
	(b)	(i)	$\frac{119.1}{45} = 2.65$		1		1	1	
		(ii)	$\frac{45}{28} = 1.61$ [A]		1		1	1	
		(iii)	$X_C = \frac{V}{I}$ used [74.1 Ω] or $X_C = \frac{1}{\omega C}$ (1) Answer = 26.2 [Hz] (1)		2		2	2	
		(iv)	$X_L = \omega L$ used with 74.1 (1) ecf Answer = 0.45 [H] (1)	1	1		2	2	
		(v)	Peak V of 168 V (1) Phase as shown (1) 		2		2	1	
		(vi)	Reactance of inductor increases or reactance of capacitor decreases or $X_L > X_C$ (1) Impedance increases because reactances don't cancel (1)		2		2		